

Unified Enterprise AI Agent

A multi-domain conversational agent that routes across HR, Support, Privacy, Finance, and Legal knowledge bases — answering, formatting, and self-checking in real time.

01

Route, don't guess

A supervisor agent classifies intent per turn — including mid-conversation topic switches — before any retrieval happens.

02

Answer only from evidence

Every response is generated strictly from retrieved chunks, then re-checked by a second model call for groundedness.

03

Shape the output, not just the answer

The same answer can flow into Markdown, Email, JSON, CSV, or XML — governed by what the requester actually needs.

5

Knowledge Domains

5

Dynamic Output Formats

System Architecture

Production vision (full diagram) vs. what runs in this prototype



Built for this prototype

- 5-domain classifier with mid-conversation switching
- Per-domain ChromaDB retrieval (sentence-transformers)
- Live faithfulness / groundedness guardrail
- 5 output formats + governance-by-prompt
- .jsonl audit log for every turn

Scoped for production roadmap

- Auth / RBAC (AD, SSO) and multi-channel intake
- Live ETL from SharePoint / Salesforce / ServiceNow
- Human-in-the-loop escalation workflow
- Full analytics dashboard + offline RAGAS suite

Tech Stack

Chosen for a reliable, dependency-light, free-tier build



Google Gemini API

LLM for classification, generation, and the live faithfulness check.
Free tier, model-agnostic fallback chain.



ChromaDB

Local vector store, one persistent collection per domain. No cloud account or hosting cost.



sentence-transformers

Local embeddings — no external embedding API dependency or added latency.



Streamlit

Single-page chat UI with multi-turn session state, styled in Kohler's brand palette.



Plain Python

No LangChain — retrieval, routing, and formatting written directly for transparent debugging.



GitHub + Streamlit Cloud

Version-controlled source, auto-redeploying public demo on every push.

Why This Approach Works

Innovation pitch

1

The AI checks its own work

After every answer, a second AI call asks: “Is this actually backed by the source documents?” If not, it's flagged — so answers stay accurate, not just confident-sounding.

2

One answer, five formats

The same answer can be turned into an email, a JSON file, or a spreadsheet — without looking anything up twice. Just the presentation changes, not the underlying facts.

3

Privacy is built into the instructions

Every agent is directly told never to reveal sensitive personal data, even if it's technically present in what it retrieved. No extra filtering tool needed — it simply refuses when asked.